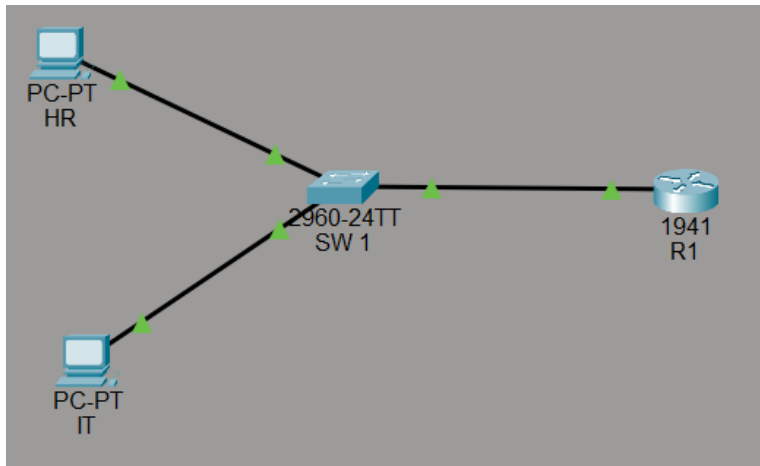


MINI PROJECT 2

Inter-VLAN Routing using Router-on-a-Stick (ROAS) + 802.1Q + ARP Flow

DIAGRAM:



DESCRIPTION:

This project explains how Inter-VLAN Routing using Router-on-a-Stick (ROAS) allows different VLANs (separate broadcast domains) to communicate with each other. It also shows how 802.1Q tagging and the ARP process work together so devices can find the router and send traffic between VLANs.

STEP 1: Create VLANs on SW 1:

Verify existing VLANs from Previous projects

STEP 2: Assign Access Ports to VLANs:

```
interface fastEthernet0/1
switchport mode access
switchport access vlan 10
exit
```

```
interface fastEthernet0/2
switchport mode access
switchport access vlan 20
exit
```

STEP 3: Configure Trunk Link (802.1Q):

```
interface gigabitEthernet0/1
switchport mode trunk
switchport trunk allowed vlan 10,20
exit
```

STEP 4: Configure IP Addressing on PCs:

PC	VLAN	IP Address	Subnet Mask
PC-HR	10	192.168.10.10/24	255.255.255.0
PC-IT	20	192.168.20.10/24	255.255.255.0

Each VLAN must use a different IP network.

STEP 5: Create Router Subinterfaces:

```
interface gigabitEthernet0/0
no shutdown
exit
```

```
interface gigabitEthernet0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
exit
```

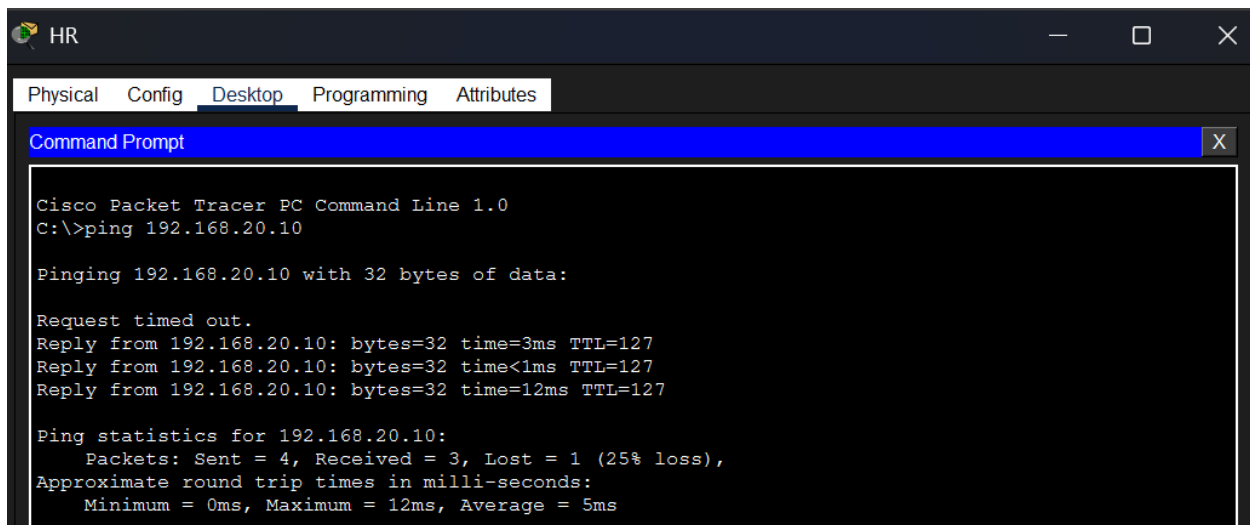
```
interface gigabitEthernet0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
exit
```

STEP 6: Verify VLAN, Interfaces & Trunk Behavior:

show vlan brief
Show ip interface brief
Show interfaces trunk

STEP 7: Ping Test & Observe ARP:

Ping from HR to IT pc:



```
HR
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.10: bytes=32 time=3ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time=12ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 5ms
```

Check ARP:

```
Router#show arp
Protocol Address Age (min) Hardware Addr Type Interface
Internet 192.168.10.10 6 0030.F2E7.153D ARPA GigabitEthernet0/0.10
Internet 192.168.20.10 6 0005.5E79.67DB ARPA GigabitEthernet0/0.20
Router#
```